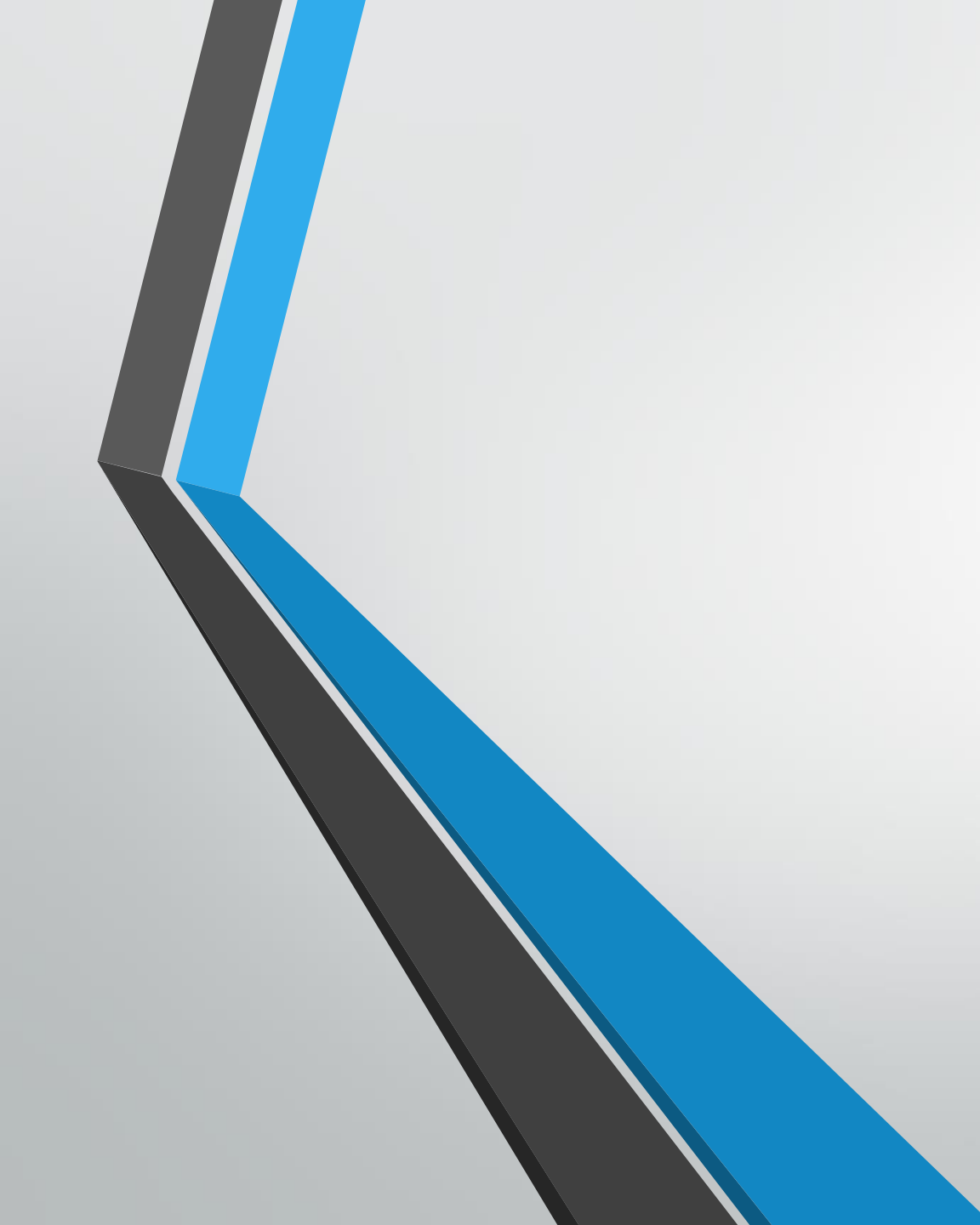




Using Hadoop

HDFS Manipulation

Pr. Ghazouani mohamed

Manipulation du contenu HDFS

There are various ways to interact with an HDFS file system, such as Java APIs, classic network protocols (HTTP, FTP, WebHDFS, HttpFS...), proprietary protocols (Amazon S3), or simply via the command line. We'll detail this last method of interaction in the rest of this section.

All commands supported by HDFS are available via the following commands:

```
$ hadoop fs
```

```
$ hdfs dfs -fs
```

An HDFS command must be called as a subcommand of the Hadoop command.

Commands Option

- **Usage:**

-ls

- **hdfs dfs -ls uri or path**

- For a file ls returns stat on the file with the following format:

permissions number_of_replicas userid groupid filesize modification_date
modification_time filename

- For a directory it returns list of its direct children as in Unix. A directory is listed as:

permissions userid groupid modification_date modification_time dirname

- Files within a directory are order by filename by default

- **Example:**

We provide an example below that lists the content of the root directory using an absolute path via a URI /.

hdfs dfs -ls /

Below we get the same result this time by using an absolute path via a hdfs://localhost:9000/ URL.

- **hdfs dfs -ls hdfs://localhost:9000/**

Commands Option

-mkdir

- **Usage:**

- **hdfs dfs -mkdir [-p] <paths>**
- Takes path uri's as argument and creates directories

- **Options:**

- The -p option behaviour is much like Unix mkdir -p, creating parent directories along the path

- **Example:**

- `hdfs dfs -mkdir /ghazouani/`
- `hdfs dfs -mkdir -p /ghazouani/mohamed/dir1`
- `hdfs dfs -mkdir -p hdfs://localhost:9000/ghazouani/hadoop/dir1`

Commands Option

-touchz

- **Usage:**
 - `hdfs dfs -touchz URI [URI ...]`
 - Create a file of zero length
- **Example:**
 - `hdfs dfs -touchz /ghazouani/fichier1.txt`

Commands Option

-put

- **Usage:**

- **hdfs dfs -put <localsrc> <dst>**
- Copy single src, or multiple srcs from local file system to the destination file system

- **Examples:**

- `hdfs dfs -put localfile5.txt /user/root/localfile6`
- `hdfs dfs -put localfile1.txt localfile2.txt /user/root/hadoopfile`

- **Usage:**

-copyFromLocal

- **hdfs dfs -copyFromLocal <localsrc> URI**
- Similar to put command.

- **Options:**

- The -f option will overwrite the destination if it already exists

For both commands **-put** and **-copyFromLocal**

- **Examples:**

- `hdfs dfs -copyFromLocal localfile3 /`

Commands Option

-get

- **Usage:**

- **hdfs dfs -get <src> <localdst>**

- Copy files to the local file system

- **Examples:**

- *hdfs dfs -get /user/root/localfile3*

- ***hdfs dfs -get /ghazouani/fichier.txt***

- *hdfs dfs -get hdfs://localhost/user/root/hadoopfile localfiles*

- ***hdfs dfs -get /ghazouani/fichier.txt Bureau***

- **Usage:**

-copyToLocal

- **hdfs dfs -copyToLocal URI <localdst>**

- Similar to get command.

- **Examples:**

- **hdfs dfs -copyToLocal /ghazouani/fichier.txt**

- *hdfs dfs -copyToLocal /user/root/localfile3 Bureau*

- *-f ? rm -f Bureau/fichier.txt && hdfs dfs -get /ghazouani/fichier.txt Bureau*

Commands Option

-moveFromLocal, -moveToLocal

- **Command: moveFromLocal**

- **Usage:**

- **hdfs dfs -moveFromLocal <localsrc> <dst>**
- Similar to put command, except that the source local src is deleted after it's copied

- **Example:**

- **hdfs dfs -moveFromLocal localfile.txt /user/root/**
- **hdfs dfs -moveFromLocal fichier1.txt /ghazouani**

- **Command: moveToLocal**

- **Usage:**

- **hdfs dfs -moveToLocal <src> <dst>**
- Displays a "Not implemented yet" message
- In practice, the data is intended to remain in HDFS, not to be used locally.
- This is why Hadoop preferred to leave the responsibility to the user to perform two separate operations: **Get + Delete**

Commands Option

-rm

- **Usage:**

- **hdfs dfs -rm [-f] [-R | -r] URI or path [URI ...]**
- Delete files specified as args

- **Options:**

- The -f option will not display a diagnostic message or modify the exit status to reflect an error if the file does not exist
- The -R option deletes the directory and any content under it recursively.
- The -r option is equivalent to -R

- **Example:**

- **hdfs dfs -mkdir /repasup**
- **hdfs dfs -touchz /repasup/fichasup.txt**
- **hdfs dfs -touchz /repasup/fichasup1.txt**
- **hdfs dfs -rm /repasup/fichasup.txt**
- **hdfs dfs -rm -r /repasup/**

Commands Option

-cp

- **Usage:**

- **hdfs dfs -cp [-f] URI or path <dest>**
- Copy files from source (HDFS) to destination (HDFS).

- **Examples:**

- **hdfs dfs -cp /user/root/localfile1 /user/root/localfile4**
- **hdfs dfs -cp /user/root/localfile1 /user/root/localfile2 /user/root/dir1**
- **hdfs dfs -cp hdfs://localhost/user/root/localfile1
hdfs://localhost/user/root/hadoopfile4**
- **hdfs dfs -cp /ghazouani/fichier.txt /result**

Commands Option

-mv

- **Usage:**

- **hdfs dfs -mv URI [URI ...] <dest>**
- Moves files from source (HDFS) to destination (HDFS)
- Allows multiple sources as well in which case the destination needs to be a directory

- **Example:**

- **hdfs dfs -mv /user/root/localfile1 /user/root/localfile5**
- **hdfs dfs -mv /ghazouani/fichier1.txt /result**

Commands Option

-cat

- **Usage:**
 - **hdfs dfs -cat URI or path**
 - Copies file content to stdout
- **Examples:**
 - `hdfs dfs -cat hdfs://localhost/user/root/localfile3`
 - `hdfs dfs -cat /ghazouani/testl.txt`

Commands Option

-tail

- **Usage:**

- **hdfs dfs -tail [-f] URI or path**
- Displays last kilobyte of the file to stdout
- The option -f stands for "**f**ollow" or "**f**orever" in this context and works very similarly to the classic tail -f command found on Unix/Linux systems.
- The -f option is primarily used to monitor log files or files that are actively being written to by a process.

- **Options:**

- The -f option will output appended data as the file grows, as in Unix

- **Example:**

- hdfs dfs -tail pathname
- hdfs dfs - tail /test1/test1.txt

Commands Option

-text

- **Usage:**
 - **hdfs dfs -text <src>**
 - Takes a source file and outputs the file in text format
- **Example:**
 - **hdfs dfs -text /user/root/filename**

Commands Option

-appendToFile

- **Usage:**

- **hdfs dfs -appendToFile <localsrc> <dst>**
- Append single src, or multiple srcs from local file system to the destination file system
- Also reads input from stdin and appends to destination file system

- **Examples:**

- **hdfs dfs -appendToFile localfile /user/hadoop/localfile2**
- **hdfs dfs -appendToFile localfile1 localfile2 /user/hadoop/hadoopfile**
- **hdfs dfs -appendToFile localfile hdfs://localhost/hadoop/localfile2**
- **hdfs dfs -appendToFile fichier.txt /poeme.txt**
- **hdfs dfs -appendToFile - /hadoop/localfile2.txt**

Commands Option

-setrep

- **Usage:**

- **hdfs dfs -setrep [-w] <numReplicas> <path>**
- Changes the replication factor of a file
- If path is a directory then the command recursively changes the replication factor of all files under the directory tree rooted at path

- **Options:**

- The -w flag requests that the command wait for the replication to complete. This can potentially take a very long time.

- **Example:**

- **hdfs dfs -setrep -w 3 /user/root/dir1**
- **hdfs dfs -setrep -w 2 /ghazouani/poeme.txt**

Commands Option

-getmerge

- **Usage:**

- **hdfs dfs -getmerge <src> <localdst> [addnl]**
- Takes a source directory and a destination file as input and concatenates files in src into the destination local file
- Optionally -nl can be set to enable adding a newline character at the end of each file

- **Example:**

- `hdfs dfs -getmerge /user/root/temp localfile`
- `hdfs dfs -getmerge /poeme.txt /ghazouani/fichier1.txt getmerge.txt`
- **`hdfs dfs -getmerge -nl /poeme.txt /ghazouani/fichier1.txt getmerge.txt`**

Commands Option

-stat

- **Usage:**

- **`hdfs dfs -stat [format] <path> ...`**
- Print statistics about the file/directory at <path> in the specified format.
- Format accepts filesize in blocks (%b), type (%F), group name of owner (%g), name (%n), block size (%o), replication (%r), user name of owner(%u), and modification date (%y, %Y). %y shows UTC date as "yyyy-MM-dd HH:mm:ss" and %Y shows milliseconds since January 1, 1970 UTC.
- If the format is not specified, %y is used by default

- **Example:**

- **`hdfs dfs -stat "%F %u:%g %b %y %r %n" /file`**

Commands Option

-du

- **Usage:**

- **hdfs dfs -du [-s] [-h] URI or path [URI ...]**
- Displays sizes of files and directories contained in the given directory or the length of a file in case its just a file

- **Options:**

- The -s option will result in an aggregate summary of file lengths being displayed, rather than the individual files.
- The -h option will format file sizes in a "human-readable" fashion (e.g 64.0m instead of 67108864)

- **Example:**

- `hdfs dfs -du /user/root/`
- `hdfs dfs -du -s hdfs://localhost/user/root/`

Commands Option

-help

- **Usage**
 - **hdfs dfs -help <command>**
 - Returns usage output
- **Example**
 - hdfs dfs -help
 - hdfs dfs -help ls
 - **hdfs dfs -help getmerge**



HDFS practical work

